

Fields and inverses

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An example of finding inverse in a prime field

We take 357 and 95, $\text{GCD}(357, 95) = 1$, 357 is not prime, but we can still find the value such that $357x + 95y = 1$.

GCD Euclidean

$$R_{-1} = 357, R_0 = 95$$

index	R_{i-2}	=	R_{i-1}	* F_i	+ R_i
1	357	=	95	*(3)	+ 72
2	95	=	72	*(1)	+ 23
3	72	=	23	*(3)	+ 3
4	23	=	3	*(7)	+ 2
5	3	=	2	*(1)	+ 1

Inverse

$$R_i = R_{i-2} * \text{COEFF}_{i-2} + R_{i-1} * \text{COEFF}_{i-1}$$

$$R_5 = R_3 * (1) + R_4 * (-F_5)$$

$$R_5 = R_2 * (-F_5) + R_3 * (1 + F_4 F_5)$$

$$R_5 = R_1 * (1 + F_4 F_5) + R_2 * (-F_3 - F_5 - F_3 F_4 F_5)$$

$$R_5 = R_0 * (-F_3 - F_5 - F_3 F_4 F_5) + R_1 * (1 + F_4 F_5 + F_2 F_3 + F_2 F_5 + F_2 F_3 F_4 F_5)$$

$$R_5 = R_{-1} * (1 + F_4 F_5 + F_2 F_3 + F_2 F_5 + F_2 F_3 F_4 F_5) + R_0 * (-F_3 - F_5 - F_3 F_4 F_5 - F_1 - F_1 F_4 F_5 - F_1 F_2 F_3 - F_1 F_2 F_5 - F_1 F_2 F_3 F_4 F_5)$$

The inverse of $R_0 \bmod R_{-1}$, or $95 \bmod 357$ is the COEFF_0 in last round $(-F_3 - F_5 - F_3 F_4 F_5 - F_1 - F_1 F_4 F_5 - F_1 F_2 F_3 - F_1 F_2 F_5 - F_1 F_2 F_3 F_4 F_5)$

$$\begin{aligned} \text{COEFF}[3] &= 1, & \text{COEFF}[4] &= -F_5 \\ \text{COEFF}[2] &= \text{COEFF}[4] + \text{COEFF}[3] * (-F_4) \\ \text{COEFF}[1] &= \text{COEFF}[3] + \text{COEFF}[2] * (-F_3) \\ \text{COEFF}[0] &= \text{COEFF}[2] + \text{COEFF}[1] * (-F_2) \end{aligned}$$

$$\frac{\text{COEFF}[-1]}{\text{COEFF}[1]} = \frac{\text{COEFF}[0] + \text{COEFF}[1] \cdot (-F1)}{\text{COEFF}[0] - F2 \cdot \text{COEFF}[1]}$$

